

Math211

Discrete Mathematics



Predicate Logic
Lecture 3

Agenda

- ▶ 1.4 Predicates and Quantifiers
- ▶ 1.5 Nested Quantifiers

Discrete Mathematics and Its Applications
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DISCRETE MATHEMATICS

1.4 Predicates and Quantifiers

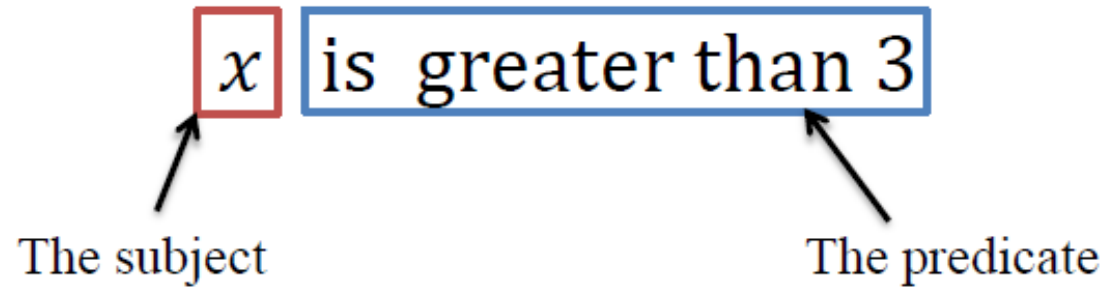
Introduction to Predicate Logic

- ▶ Propositional logic, cannot adequately express the meaning of all statements in mathematics and in natural language. For example, suppose that we know that
 - ▶ **“Every computer connected to the university network is functioning properly.”**
- ▶ No rules of propositional logic allow us to conclude the truth of the statement
 - ▶ **“MATH3 is functioning properly,”** where MATH3 is one of the computers connected to the university network.

Introduction to Predicate Logic

- ▶ In this section we will introduce a more powerful type of logic called predicate logic. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects.
- ▶ To understand predicate logic, we first need to introduce the concept of a predicate.
- ▶ Afterward, we will introduce the notion of quantifiers, which enable us to reason with statements that assert that a certain property holds for all objects of a certain type and with statements that assert the existence of an object with a particular property.

Predicate Logic



- ▶ We can denote the predicate (condition) "*is greater than 3*" and x is the variable (subject). The statement " x is greater than 3" by $P(x)$ (propositional function), where P denotes
- ▶ The statement $P(x)$ is also said to be the value of the propositional function P at x .
- ▶ Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Predicate Logic

The predicate logic include:

- ▶ **Variables:** x, y, z , these are the subjects of the statements
- ▶ **Predicate (P):** a property the variable can have
- ▶ **Propositional function $P(x)$:** is the result of applying a predicate P to an object x is the
When the variables in a propositional function $P(x)$ are assigned values, the resulting statement becomes a proposition with a certain truth value.
- ▶ **Domain or universe:** represents range of values of a variable
- ▶ **Quantifiers:** expresses extent to which a predicate is true over a range of elements.

Predicate Logic

Let $P(x)$ denotes the statement “ x is greater than 3”. What are the truth values of $P(4)$ and $P(2)$

- We obtain the statement $P(4)$ by setting $x = 4$ in the statement “ $x > 3$.” Hence, $P(4)$, which is the statement “ $4 > 3$,” is **true**.
- **However**, $P(2)$, which is the statement “ $2 > 3$,” is **false**.

Predicate Logic

Let $P(x)$ denotes the statement “ x is an even number”. What are the truth values of $P(5)$ and $P(8)$

- We obtain the statement $P(5)$ by setting $x = 5$ in the statement “5 is an even number.” Hence, $P(5)$, is **false**.
- **However**, $P(8)$, which is the statement “8 is an even number,” is **true**.

Predicate Logic

- ▶ Convention:
 - ▶ **Lowercase** variables $x, y, z...$ denote **objects**/entities.
 - ▶ **Uppercase** variables $P, Q, R...$ denote propositional functions (**predicates**).
- ▶ A predicate is a **propositional** function of **one or more** variables in a given domain.

Example

Let $Q(x, y)$ denote the statement " $x = y + 3$ ". What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

F

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- ▶ When the *variables* in a propositional function are *assigned values*, the resulting statement becomes a *proposition* with a certain truth value.

Example

1. Let $P(x)$ denote the statement “ $x \leq 4$.” What are the truth values?
 - a) $P(0)$
 - b) $P(4)$
 - c) $P(6)$
2. Let $P(x)$ be the statement “the word x contains the letter a .” What are the truth values?
 - a) $P(\text{orange})$
 - b) $P(\text{lemon})$
 - c) $P(\text{true})$
 - d) $P(\text{false})$

Example

1. Let $P(x)$ denote the statement " $x \leq 4$." What are the truth values?
a) $P(0)$ **T** b) $P(4)$ **T** c) $P(6)$ **F**
2. Let $P(x)$ be the statement "the word x contains the letter a ." What are the truth values?
a) $P(\text{orange})$ **T** b) $P(\text{lemon})$ **F**
c) $P(\text{true})$ **F** d) $P(\text{false})$ **T**

Quantifiers

Quantifiers

A propositional function $P(x)$ is not a proposition until it has a truth value.

- ▶ Up to this point, we could only do this by **assigning a value** to our variable.
 - ▶ Example: If $P(x)$ represents " $x > 0$ ", find the truth value for $P(4)$
- ▶ There is another way to turn a propositional function into a proposition **using a quantifier**.

Quantifiers

- ▶ In English, the words **all**, **some**, **many**, **none**, and **few** are used in quantifications.
- ▶ **Quantification** expresses the extent to which a predicate is true over a range of elements.
- ▶ **Quantification** can be used to create a proposition from a propositional function.
 - ▶ It describes the values of a variable that make the predicate true.
 - ▶ **Domain** or **universe**: range of values of a variable

Two Popular Quantifiers

1. **Universal:** $\forall x P(x)$ – “P(x) is true for all x in the domain” “for all values of x in the domain, P(x) is true” [x has the property (predicate)]
 2. **Existential:** $\exists x P(x)$ – “P(x) is true for some values of x in the domain,” or “there exists a value of x (that is, 1 or more) such that P(x) is TRUE” [x has the property (predicate)]
- Either is meaningless if the domain is not known/specified.

Truth of Quantifiers

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

$P(x)$ denote a predicate in the domain $U = \{1, 2, 3\}$

$$\forall x P(x) \Leftrightarrow P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \Leftrightarrow P(1) \vee P(2) \vee P(3)$$

Infinite Sets

► Symbols for some special **infinite sets**:

N = {0, 1, 2, ...} The **N**atural numbers.

Z = {..., -2, -1, 0, 1, 2, ...} The **I**ntegers.

Z⁺ = {1, 2, 3, 4, ...} The positive **I**ntegers.

R = The “**R**eal” numbers, such as 3.5, 3.1, 1.2, 5, 4,

Example

Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution: Because $P(x)$ is true for all real numbers x , the quantification

$$\forall x P(x)$$

is true.



Example

Let $Q(x)$ be the statement “ $x < 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. Thus

$$\forall x Q(x)$$

is false.

Example

What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement “ $x^2 < 10$ ” and the domain consists of the positive integers not exceeding 4?

Solution: The statement $\forall x P(x)$ is the same as the conjunction

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4),$$

because the domain consists of the integers 1, 2, 3, and 4. Because $P(4)$, which is the statement “ $4^2 < 10$,” is false, it follows that $\forall x P(x)$ is false. 

Example

What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution: The universal quantification $\forall x(x^2 \geq x)$, where the domain consists of all real numbers, is false. For example, $(\frac{1}{2})^2 \not\geq \frac{1}{2}$. Note that $x^2 \geq x$ if and only if $x^2 - x = x(x - 1) \geq 0$. Consequently, $x^2 \geq x$ if and only if $x \leq 0$ or $x \geq 1$. It follows that $\forall x(x^2 \geq x)$ is false if the domain consists of all real numbers (because the inequality is false for all real numbers x with $0 < x < 1$). However, if the domain consists of the integers, $\forall x(x^2 \geq x)$ is true, because there are no integers x with $0 < x < 1$.

Example

Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution: Because “ $x > 3$ ” is sometimes true—for instance, when $x = 4$ —the existential quantification of $P(x)$, which is $\exists x P(x)$, is true. ◀

Example

What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement “ $x^2 > 10$ ” and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists x P(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4).$$

Because $P(4)$, which is the statement “ $4^2 > 10$,” is true, it follows that $\exists x P(x)$ is true.



Example

Let $P(x)$ be the statement “ $x = x^2$.” If the domain consists of the integers, what are the truth values?

- | | | |
|------------|---------------------|---------------------|
| a) $P(0)$ | b) $P(1)$ | c) $P(2)$ |
| d) $P(-1)$ | e) $\exists x P(x)$ | f) $\forall x P(x)$ |

Let $P(x)$ be the statement “ $x = x^2$.” If the domain consists of the integers, what are the truth values?

- | | | |
|---------------------|------------------------------|------------------------------|
| a) $P(0)$ T | b) $P(1)$ T | c) $P(2)$ F |
| d) $P(-1)$ F | e) $\exists x P(x)$ T | f) $\forall x P(x)$ F |

The Universal Quantifier $\forall$

► Example:

Let $P(x)$ denote the statement " x has taken a course in calculus" and the domain of x be the students in the class.

Then the universal quantification of $P(x)$, $\forall x P(x)$, is the proposition:

- "Every student in the class has taken a course in calculus"
- "All students in the class have taken a course in calculus"
- "Each student in the class has taken a course in calculus"

The Existential Quantifier $\exists$

► Example:

Let $P(x)$ denote the statement " x has taken a course in calculus" and the domain of x be the students in the class.

Then the universal quantification of $P(x)$, $\exists x P(x)$, is the proposition:

- "some student in the class has taken a course in calculus"
- "There is a student in the class has taken a course in calculus"
- "At least one student in the class has taken a course in calculus"

The Universal Quantifier $\forall$

► Example:

Let $P(x)$ denote the statement “ x is full.”

Let the domain of x be parking spaces at Nile University.

Then the *universal quantification* of $P(x)$, $\forall x P(x)$, is the *proposition*:

- “All parking spaces at NU are full.”
- *i.e.*, “Every parking space at NU is full.”
- *i.e.*, “For each parking space at NU, that space is full.”

The Existential Quantifier $\exists$

► Example:

Let $P(x)$ denote the statement “ x is full.”

Let the domain of x be parking spaces at Nile University.

Then the *existential quantification* of $P(x)$, $\exists x P(x)$, is the *proposition*:

- “Some parking space at NU is full.”
- “There is a parking space at NU that is full.”
- “At least one parking space at NU is full.”

Precedence of Quantifiers and Logical Operators

Operator	Precedence
$\forall, \exists$	1
$\neg$	2
$\wedge$	3
$\vee$	4
$\rightarrow$	5
$\leftrightarrow$	6

Precedence of Quantifiers

- ▶ $\forall$ $\exists$ have higher precedence than operators from Propositional Logic; so
 - ▶ $\forall x P(x) \vee Q(x)$ is not logically equivalent to $\forall x (P(x) \vee Q(x))$
- ▶ Logical Equivalence: $P \equiv Q$ iff they have same truth value no matter which **domain** is used and no matter which **predicates** are assigned to predicate variables.

Free and Bound Variables

- ▶ An expression like $P(x)$ is said to have a *free variable* x (meaning, x is undefined).
- ▶ A quantifier (either $\forall$ or $\exists$) *operates* on an expression having one or more free variables, and *binds* one or more of those variables, to produce an expression having one or more *bound variables*.

Remember

A *propositional function (predicate)* is not a *proposition* until *all* variables have been bound either by quantification or assignment of a value!

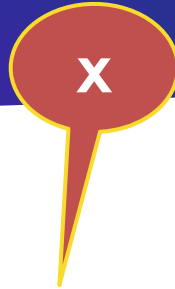
Binding Variables

- ▶ The occurrence of a variable x is **bound** iff a value is assigned to x or when a quantifier is used on x . Otherwise, it is called **free**.

$$\exists x Q(x, y)$$

$$\exists x (P(x) \vee Q(x)) \wedge \forall x (R(x))$$

Example of Binding

yx

- ▶ $P(x,y)$ has 2 free variables, x and y .
- ▶ $\forall x P(x,y)$ has 1 free variable, and one bound variable. [Which is which?]
- ▶ “ $P(x)$, where $x=3$ ” is another way to bind x .
- ▶ An expression with zero free variables is an **actual proposition**.
- ▶ An expression with one or more **free** variables **is still only a predicate**: $\forall x P(x,y)$

Examples

- ▶ Let $U = \mathbb{Z}$, the integers = $\{\dots -2, -1, 0, 1, 2, 3, \dots\}$
 - ▶ $P(x): x > 0$ is the predicate. It has no truth value until the variable x is bound.
- ▶ Examples of propositions where x is assigned a value:
 - ▶ $P(-3)$ is false,
 - ▶ $P(0)$ is false,
 - ▶ $P(3)$ is true.
- ▶ The collection of integers for which $P(x)$ is true are the positive integers.

Examples

- ▶ $P(y) \vee \neg P(0)$ is not a proposition.
 - ▶ The variable y has not been bound.
 - ▶ However, $P(3) \vee \neg P(0)$ is a proposition which is true.

Examples

Let R be the three-variable predicate $R(x, y, z)$:

$$x + y = z$$

► Find the truth value of

$$R(2, -1, 5), R(3, 4, 7), R(x, 3, z)$$

F

T

hasn't a truth value

Negation of Quantifiers

► Example:

Let $P(x)$ denote the statement " x has taken a course in calculus" and the domain of x be the students in the class.

Then the universal quantification of $P(x)$, $\forall x P(x)$, is the proposition:

► "Every student in the class has taken a course in calculus"

The negation of this statement is

"There is at least one student in your class who has not taken a course in calculus"

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

Negation of Quantifiers

► Example:

Let $P(x)$ denote the statement " x has taken a course in calculus" and the domain of x be the students in the class.

Then the existential quantification of $P(x)$, $\exists x P(x)$, is the proposition:

- "At least one student in your class has taken a course in calculus"

The negation of this statement is

"Every student in this class has not taken calculus"

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Negation of Quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

- **Careful:** The negation of “Every student in the class has taken a course in calculus” is NOT “No student in the class has taken a course in calculus”!

Negation of Quantifiers

- ▶ Distributing a negation operator across a quantifier changes a universal to an existential and vice versa.

$$\begin{aligned}\neg(\forall x P(x)) &\Leftrightarrow \neg(P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)) \\ &\Leftrightarrow \neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n) \\ &\Leftrightarrow \exists x \neg P(x)\end{aligned}$$

De Morgan's Laws for Quantifiers

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Review Questions



Problem 1

- ▶ $P(x)$: “x can speak Russian”
 - ▶ $Q(x)$: “x knows the computer language C++.”
 - ▶ Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives.
 - ▶ **The domain for quantifiers consists of all students at your school.**
- a) There is a student at your school who can speak Russian and who knows C++.

$$\exists x(P(x) \wedge Q(x))$$

Problem 1

- ▶ $P(x)$ be the statement “ x can speak Russian”
- ▶ $Q(x)$ be the statement “ x knows the computer language C++.”
- b) There is a student at your school who can speak Russian but who doesn't know C++.

$$\exists x(P(x) \wedge \neg Q(x))$$

- c) Every student at your school either can speak Russian or know C++.

$$\forall x(P(x) \vee Q(x))$$

- d) No student at your school can speak Russian or know C++.

$$\neg \exists x(P(x) \vee Q(x)) \quad \forall x \neg(P(x) \vee Q(x)) \quad \forall x (\neg P(x) \wedge \neg Q(x))$$

Problem 2

$P(x)$ = “x is a professor” $Q(x)$ = “x is ignorant” $R(x)$ = “x is vain”

Express the following statements using quantifiers, logical connectives, and $P(x)$, $Q(x)$, and $R(x)$, **where the universe of discourse (Domain) is the set of all people.**

1. No professors are ignorant

For every person, if that person is a professor, then that person is not ignorant.

$$\forall x(P(x) \rightarrow \neg Q(x))$$

2. All ignorant people are vain

If a person is ignorant, then they must also be vain.

$$\forall x(Q(x) \rightarrow R(x))$$

3. No professors are vain

For every person, if that person is a professor, then that person is not ignorant.

$$\forall x(P(x) \rightarrow \neg R(x))$$

Problem 3

Every Student in this class studied discrete math.

Domain: All students in this class $D(x)$: x study discrete math

$$\forall x D(x)$$

Domain: All people $S(x)$: x is a student in this class $D(x)$: x study discrete math

$$\forall x (S(x) \rightarrow D(x))$$

Domain: All people $S(x)$: x is a student $C(x)$: x in this class $D(x)$: x study discrete math

$$\forall x (S(x) \wedge C(x) \rightarrow D(x))$$

Domain: All Student $C(x)$: x is in this class $D(x)$: x study discrete math

$$\forall x (C(x) \rightarrow D(x))$$

Problem 4

Some Students in this class studied discrete math.

Domain: All students in this class $D(x)$: x study discrete math

$$\exists x D(x)$$

Domain: All people $S(x)$: x is a student in this class $D(x)$: x study discrete math

$$\exists x (S(x) \wedge D(x))$$

We can't use implies operator for the existential quantifier

Domain: All people $S(x)$: x is a student $C(x)$: x in this class $D(x)$: x study discrete math

$$\exists x ((S(x) \wedge C(x)) \wedge D(x))$$

Domain: All Student $C(x)$: x is in this class $D(x)$: x study discrete math

$$\exists x (C(x) \wedge D(x))$$



Problem 5

Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives

Everything is in the correct place and in excellent condition

Domain:

The domain of discourse is likely "all items" or "things"

Predicates:

$P(x)$: x is in the correct place

$C(x)$: x is in excellent condition.

$\forall x(P(x) \wedge C(x))$



Problem 6

Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives

Nothing is in the correct place and is in excellent condition.

Domain:

The domain of discourse is likely "all items" or "things"

Predicates:

$P(x)$: x is in the correct place

$C(x)$: x is in excellent condition.

$$\neg \exists x (P(x) \wedge C(x)) \equiv \forall x \neg (P(x) \wedge C(x))$$



Problem 7

- ▶ What are the negations of the statements “There is an honest politician”?

Solution:

- ▶ $H(x)$ denote “ x is honest.”
- ▶ “There is an honest politician” is represented by $\exists x H(x)$, where the domain consists of all politicians.
- ▶ Negation of this statement is $\neg \exists x H(x) \equiv \forall x \neg H(x)$.

Problem 8

What are the negations of the statements “All Americans eat cheeseburgers”?

- ▶ Let $C(x)$ denote “ x eats cheeseburgers.”
- ▶ Then “All Americans eat cheeseburgers” is represented by $\forall x C(x)$, where the domain consists of all Americans.
- ▶ The negation of this statement is $\neg \forall x C(x)$, which is equivalent to $\exists x \neg C(x)$.
- ▶ This negation can be expressed in several different ways, including “Some American does not eat cheeseburgers” and “There is an American who does not eat cheeseburgers.”

Next Lecture

See you next lecture

